

# Research on Driving Factors of COVID-19 Protective Behaviours in Australia

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## 1 Declaration

In submitting this work I am indicating that I have read the University’s Academic Integrity Policy. I declare that all material in this assessment is my own work except where there is clear acknowledgement and reference to the work of others. I give permission for this work to be reproduced and submitted to other academic staff for educational purposes.

## 2 Abstract

During the COVID-19 pandemic, personal protective behaviour, especially wearing masks, played a crucial role in controlling of the infection. Although public health policies such as mask mandates significantly changed people’s compliance, understanding the behavioural driving factors behind protective behaviours are still important. These driving factors are complex and dynamically influenced by macro pandemic factors such as vaccination rates and confirmed cases. This study investigates the determinants of COVID-19 protective behaviours in Australia through a two-phase machine learning framework. In the first phase, this study reproduced Ryan et al.’s (Ryan et al. 2025, 8–15) research, using random forest and XGBoost to classify and predict the protective behaviours before and after the mask mandates. The evaluation results showed that prior to the implementation of the mandates, people’s protective behaviour was mainly driven by existing protective habits and frequency of non-household contact. After the mandates, the passage of time (week number) and an individual’s willingness to isolate became the main predictive factors. To address the structural limitations of traditional tree models in capturing high-order and nonlinear variable interactions, the upcoming second phase proposes the construction of a Multi-layer Perceptron (MLP) architecture and the integration of state-level weekly vaccination rates and confirmed cases. In addition, to overcome the ‘black box’ characteristics of neural networks, this study introduced SHAP (SHapley Additive exPlanations) values to quantify feature contributions.

### 3 Introduction

During the COVID-19 pandemic, individual protective behaviours played a crucial role in controlling the spread of the virus (Guo, Xiang, and Wang 2023, 1–2). Among all protective behaviours, mask wearing was initially recognized as one of the most effective measures [Eikenberry et al. (2020), p.11-13). To cope with future pandemics, understanding the drivers of protective behaviours, particularly mask wearing, is essential for the design of public health policies.

There were many studies investigating why people wear masks from various perspectives (Ryan et al. 2025, 2). Among these, mandated policy is one of the most important factors, which may lead to a sharp increase in the use of masks [Haischer et al. (2020), p.4-9). Recently, Ryan et al. (Ryan et al. 2025, 8–13) investigated the role of face mask mandates in influencing the drivers of protective health behaviours in Australia. However, some key factors have not been considered. For example, the introduction of vaccination may alter individuals’ risk perception and behavioural responses (De Nicolò, Villari, and De Vito 2022, 2). There is a need to understand how different external factors drive behaviour change to inform future public health policy-making.

In order to identify these complex relationships, appropriate modelling approaches are needed. Most existing research relies on traditional statistical or tree-based models. Although tree-based models often perform strongly on tabular data, they may be limited in fully capturing complex interactions (Rana et al. 2023, 3). However, research also indicates that for most tabular data, GBDTs (gradient-boosted decision trees) outperform neural networks (NNs). In fact, the comparative performance of these two types of models on tabular data remains a subject of significant academic debate, as results often vary significantly across different datasets. Upon reviewing the literature, there is no existing research applying deep learning to similar datasets within this field.

To address these research gaps, this study integrated vaccination data and COVID-19 case numbers, and will apply deep learning method to find out whether it can capture complex relationships and identify the key drivers of protective behaviours, especially mask wearing.

For the whole study, the primary research questions are:

- How do vaccination rates and COVID-19 case numbers influence the uptake of protective behaviours in Australia?
- How does the performance of deep learning models compare to traditional statistical or tree-based models in predicting protective behaviours?

## 4 Background

## 5 Methods

This section details the experimental design and implementation methods of the study.

### 5.1 Data Sources

To explore the research questions, this study used four datasets. This section introduces the sources and contents of the data. For detailed processing procedures, please refer to the section 3.2.

#### 5.1.1 YouGov Behaviour Tracker Dataset

The Imperial College London YouGov Behaviour Tracker dataset (Jones, Imperial College London Big Data Analytical Unit, and YouGov Plc 2020) contains the self-reported survey responses of individuals from 29 countries regarding protective health behaviours against COVID-19 between 2020 and 2022. The survey included demographic and psychological factors (such as gender and age), protective behaviours (such as mask wearing and social distancing), comorbidities, and more features. As this study focuses on the situation in Australia, only the Australian subset of this dataset is used.

#### 5.1.2 Oxford COVID-19 Government Response Tracker Dataset

The Oxford COVID-19 Government Response Tracker dataset (Hale et al. 2021) records the state level response policies of governments around the world aimed at preventing the spread of COVID-19 from March 2020 to January 2023. The recorded policies include containment and closure, economic, healthcare, vaccination, and other general policies. This dataset is used for the mandate period split.

#### 5.1.3 Pandemic Case Data

Reporting of COVID-19 cases varied across each state and territory in Australia. Hence, this study uses a consolidated dataset (covid19data.com.au 2023a) for COVID-19 case numbers, which compiles and verifies COVID-19 case data from Australian federal, state, and territory health departments. The data records daily confirmed cases at state and territory level granularity, covering the period from January 2020 to August 2023. It is only used in the extension phase.

#### 5.1.4 Vaccination Data

Since the Australian government has not released detailed vaccination data for each state on a weekly basis, this study relies on a vaccine dataset compiled by a third-party team (covid19data.com.au 2023b) to match the timing and locations of the YouGov survey. This dataset collects weekly vaccination totals for each state and territory from official health websites, covering January to August 2022. It is only used in the extension phase.

### 5.2 Data Cleaning, Variable Construction, and Data Preprocessing

This study preprocessed the datasets based on the methodology of Ryan et al. (Ryan et al. 2025, 3–6) to meet the input requirements of the models. The specific feature conversion and cleaning steps are divided into eight core steps.

### 5.2.1 Government Mandates

During the COVID-19 pandemic, Australia experienced a variety of enforced orders. Due to the different response measures among local governments, the experience of COVID-19 in each state varied significantly. Therefore, the start time and enforcement of mask mandates varied greatly between different states. In order to evaluate the actual effectiveness of government mandates, this study aligned the data based on the daily mask wearing policy index (0 - No policy' to 4 -Required at all times' ) from the OxCGRT dataset (Hale et al. 2021).

Following Ryan et al.'s (Ryan et al. 2025, 3) method, and considering the varying levels and effects of policy implementation, this study calculated the 14-day rolling average of each state's index. When the rolling average of a certain state reached 3 for the first time, it was considered that the jurisdiction had actually entered a stable period of mandate intervention. Then, the completion time of the individual survey was compared to their regions, and a variable `during_mandate` was generated for each observation.

### 5.2.2 Removal of Features with High Missingness

For the YouGov dataset, due to multiple item changes during the tracking period of the survey in Australia, some questions were only asked within an extremely short window of time. To prevent the loss of massive amounts of valid samples during the cleaning phase due to high missingness in a few features, this study counted the number of missing values for each feature, and same as Ryan et al.'s (Ryan et al. 2025, 3) study, set 10,781 as the threshold. Columns with missing values greater than this threshold were considered not in the survey for a long period and were dropped from the dataset.

### 5.2.3 Repairing Data Gaps Caused by Survey Changes

During the data exploration process, it was found that YouGov modified the informed consent terms of the survey between February 10 and October 18, 2021. This led to a significant structural data gap in health related questions during this period. Deleting these records would mean losing eight months of important data. To this end, this study designed an experiment to fill in missing values, comparing the performance of the model fitted by string filling, retaining `np.nan`, and directly deleting entire rows after processing systematic missing data in this time period, and selecting the appropriate method. All three data sets were trained and tested on XGBoost, because it is the only model selected in this study that can handle `np.nan`.

As shown in the figure 1, the experimental results indicate that there is little difference in the performance of the three methods on the training set. Subsequently, this study validated their performance on the test set, and the results were all the same.

As other models in this framework fail to deal with missing values, and deleting these features directly would impair model performance. Finally, this study located this time window using a date mask and filled in these missing values as independent 'N/A' categories, thus preserving valid samples.

### 5.2.4 Processing of Predictive Variables

The core prediction targets are self protection behaviour and mask wearing. The original survey records the frequency of respondents performing a certain behaviour in the past 7 days. This study

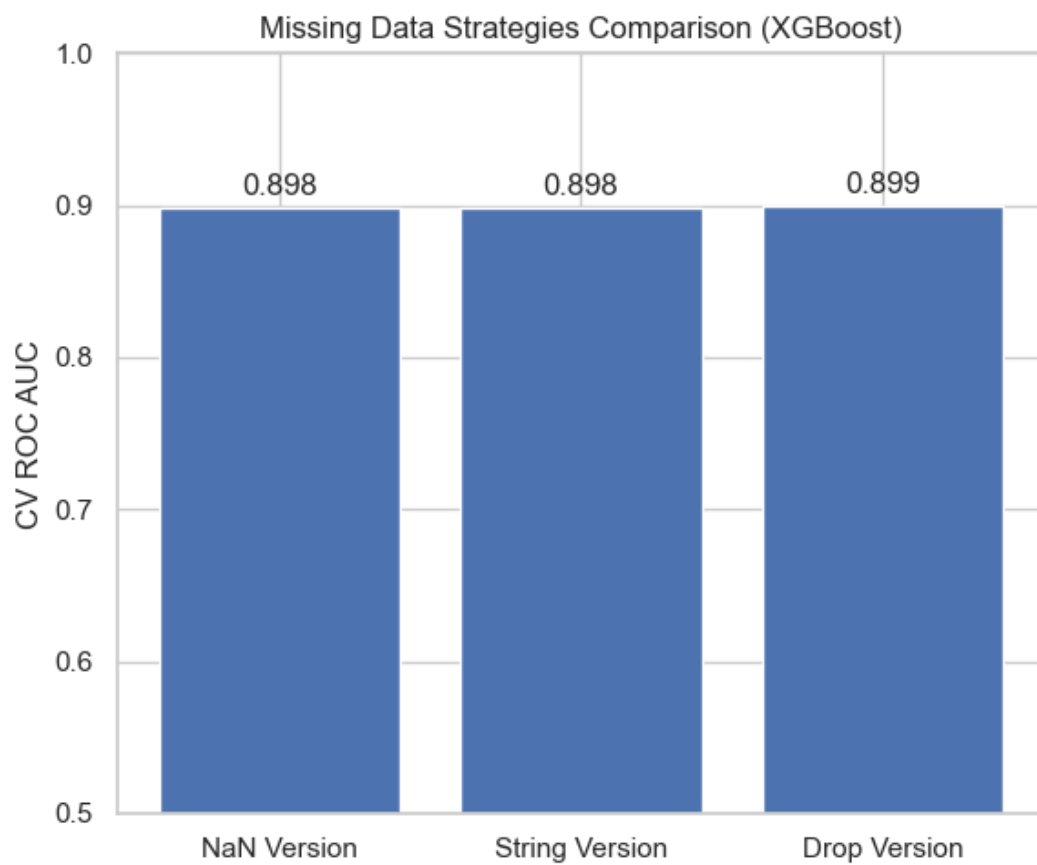


Figure 1: Performance comparison of three imputation methods based on training set cross-validation.

first constructed a dictionary and measured responses on a five-point scale (from 1 - **not at all** to 5 - **always**). Then, features were divided into three logical branches:

- **Mask wearing behaviours:** There are four specific scenarios related to masks ('Worn a face mask outside your home', 'Worn a face mask inside a grocery store/supermarket', 'Worn a face mask inside a clothing/footwear shop', and 'Worn a face mask on public transportation'). **Mask wearing behaviour** was defined as the median of all scenarios and named **face\_mask\_scale**. After that, 4 (i.e. **frequently**) was used as the compliance threshold to binarise this feature into a classification label of Yes/No, ensuring a consistent binary outcome definition with the methodology of Ryan et al. (Ryan et al. 2025, 4).
- **Protective behaviours:** All self protection scenario scores with the prefix **i12\_** were extracted, including other protective measures from the survey, such as social distancing, hand washing, and coughing covering. Their overall median was calculated as **protective\_behaviour\_scale**. The same threshold of 4 or above was used to binarise this feature into **protective\_behaviour\_bi**.
- **Protective features other than masks:** Ryan et al. (Ryan et al. 2025, 10–12) showed that a person's protective behaviours could strongly predict whether they would wear a mask. To avoid confounding effects from respondents' other hygiene habits when predicting mask behaviour, this study specifically excluded the four mask scenarios mentioned above, calculating the median of all remaining protective behaviours as a continuous feature **protective\_behaviour\_nomask\_scale**.

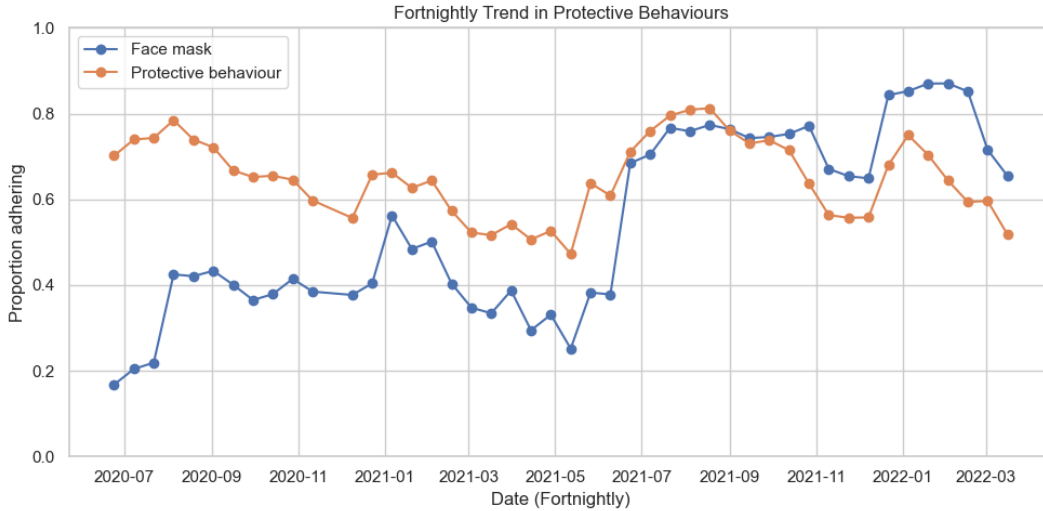


Figure 2: Public compliance between July 2020 and March 2022.

The chart 2 shows the changes in people's protective behaviour from 2020 to 2022 after cleaning and preprocessing. Initially, the protective behaviours other than masks (orange line) was higher than the proportion of mask wearing (blue line). However, in mid-2021, both classes showed a significant increase. In the later stages, the proportion of people wearing masks exceeded other protective behaviours, reaching a high point of nearly 90%. Both variables exhibit similar temporal trends.

- Other protective predictors: Additionally, this study added three other self-protective features. The first is the average number of non-household contacts `i2_health`, which was retained as a continuous numerical variable. The remaining two are subjective attitude features: willingness to isolate if unwell `i9_health` and willingness to isolate if directed to do so by a healthcare professional `i11_health`. These two subjective attitudes were recorded as categorical responses, and transformed into dummy variables prior to modelling.

### 5.2.5 Dimensionality Reduction for Sparse Comorbidity Features

In the YUGOV data, comorbidities were scattered among 13 binary options (from `d1_health_1` to `d1_health_99`), making the data extremely sparse. To improve model efficiency, this study removed all specific disease types and made them into a single variable called `d1_comorbidities`. Respondents who selected ‘None of these’ were recorded as ‘No’, ‘Prefer not to say’ was recorded as ‘Prefer\_not\_to\_say’, records within the consent gap window were imputed as ‘N/A’, and any other confirmed diseases were grouped as ‘Yes’.

### 5.2.6 Processing of Demographic Variables

The demographic variables used in this analysis include age, gender (male/female), state (one of the eight states or territories in Australia), employment status, and household size. When dealing with household size, the raw data exhibited a clear long-tail distribution, mixed with text responses. To reduce the effect of extreme values on models, this study assigned a value of 8 to all samples answering ‘8 people or more’. Invalid samples answering ‘I don’t know’ or ‘I don’t want to disclose’ were directly deleted at the end of the cleaning process.

### 5.2.7 Processing of Perception, Trust, and Mental Health Variables

In addition to demographic and behavioural data, this study included subjective variables such as perception of illness threat, trust in government, mental health, and wellbeing.

For perception of illness threat (perceived susceptibility and severity), the original responses were measured on a 7-point scale mixed with text (e.g., 7 - ‘Agree’ and 1 - ‘Disagree’). This study cleaned the text characters and converted them into continuous variables. Wellbeing (Cantril Ladder), measured on a 10-point scale, was directly retained as a continuous numerical feature. Finally, for unordered categorical features such as trust in government, PHQ-4 mental health features, and comorbidities, dummy coding was applied with `drop_first = True` to remove the baseline group and prevent multicollinearity.

### 5.2.8 Vaccination and COVID-19 Case Numbers

This study used the `endtime` of the respondents’ survey completion and their `state` value to merge the individual survey records with the corresponding state’s vaccination rates and daily new cases.



## 5.3 Resampling and Validation Pipeline

### 5.3.1 Class Imbalance and Resampling

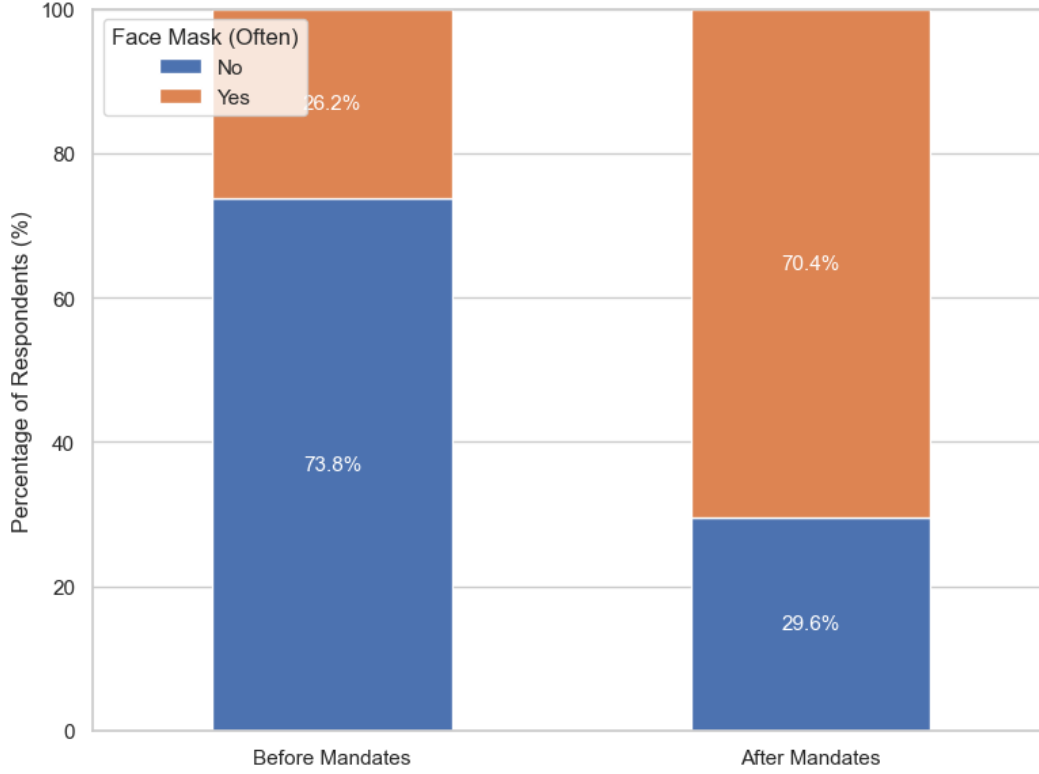


Figure 3: The proportion of respondents frequently wearing face masks before and after the implementation of mask mandates.

As shown in 3, the orange bars represent individuals who frequently wore face masks (Yes), whereas the blue bars represent those who did not (No). Before the mandates, the proportion of people who did not wear masks was as high as 73.8%, while those who frequently wore masks accounted for only 26.2%, demonstrating an extreme class imbalance. After mandates, this proportion reversed (70.4% wearing vs. 29.6% not wearing). If such imbalanced data is directly used for training, machine learning models may inherently predict the majority class, resulting in low capture ability for the minority class. To overcome this, this study introduced the random over sampler technique to deal with class imbalance.

### 5.3.2 Data Splitting and Cross Validation

After completing data cleaning and feature engineering, the data was split into training and testing sets in an 80:20 ratio, stratified by the value of `during_mandate`. To evaluate the effectiveness of policies across multiple dimensions and ensure the reliability of model training, this study constructed a specific data splitting and cross-validation mechanism.

In terms of validation strategy during training, a single validation set is highly susceptible to the randomness of the initial data split. To prevent overfitting during model training, this study implemented a 5-fold cross validation. This mechanism ensures a robust and unbiased evaluation of model performance (Allgaier and Pryss 2024, 1).

Figure 4 shows a simple demonstration of data splitting and cross validation process in this study. Initially, the dataset set was divided into a ‘training set’ and a ‘testing set’ in an 80:20 ratio. Then, a 5-fold cross validation was performed within the ‘training set’ (which was split into 5 equal parts) to find the best hyperparameter combination for the model. During each cross validation fold, one part was set aside as a ‘Validation Fold’ to verify the actual effectiveness of the parameters under this fold. This process was repeated 5 times, and the final result was the average of the five validation results.

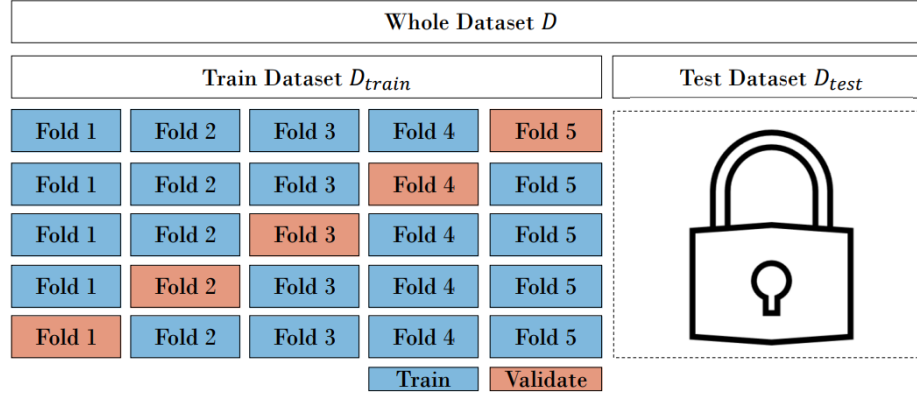


Figure 4: Schematic illustration of a 5-fold cross-validation framework. Source paper: <https://www.mdpi.com/2504-4990/6/2/65>

## 5.4 Models

### 5.4.1 Logistic Regression

### 5.4.2 Classification Trees

### 5.4.3 XGBoost

### 5.4.4 MLP Model

## 5.5 Model Evaluation and Interpretation

### 5.5.1 Evaluation Metrics

### 5.5.2 Model Interpretation

## **6 Results**

his section demonstrates and explains the performance of the models, the results of model selection, and the comparison of key predictors.

### **6.1 Model Performance and Selection**

### **6.2 Comparison of Key Predictors Before and After Mandate Periods**

#### **6.2.1 Face Mask Wearing**

#### **6.2.2 General Protective Behaviours**

## 7 Discussion

## 8 References

- Allgaier, Johannes, and Rüdiger Pryss. 2024. “Cross-Validation Visualized: A Narrative Guide to Advanced Methods.” *Machine Learning and Knowledge Extraction* 6 (2): 1378–88. <https://doi.org/10.3390/make6020065>.
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- Ryan, Michael, Jingjing Ye, Julian Sexton, Roslyn I. Hickson, and Emily Brindal. 2025. “Face Mask Mandates Alter Major Determinants of Adherence to Protective Health Behaviours in Australia.” *Royal Society Open Science* 12 (3): 241941. <https://doi.org/10.1098/rsos.241941>.

We can type equations in our new environment:

$$E\left[\frac{1}{X}\right] \neq \frac{1}{E[X]} \quad (1)$$

You can see that the text is indented and so it is clearly separated from the normal text because the next line is just normal text.

Here is the normal text outside our Definition environment. Now, how about a theorem?

### Theorem (KnuthTextbook?)

This is a theorem that we need. It comes from the earlier work cited in the title of this

environment, and we include it here so that the reader has all the information that they need to see what we are doing. As we did above, you can have dot points in your new environment.

1. first result in the theorem
2. second result in the theorem

Now an equation in the theorem

$$b = a + 2 \tag{2}$$

This text is after the theorem.

## 8.1 Algorithms

You will very likely want to describe some algorithms or processes that you followed in your project. For that, you could use the same structure as above.

### Model algorithm

- Step 1: Define the initial values of the parameters  $n, a, c_1, its$
- Step 2: Generate all the random points that we will sample from.
- Step 3: Sample  $n$  points from the population.
  - Repeat this  $its$  times.

## 8.2 Something fancy

If you want to really separate something from the main text, you could try putting a box around it with the following code.

### Theorem (KnuthTextbook?)

This is the same theorem as above, but now in a fancy box!

## 8.3 Maths equations

You can see above that we've got a couple of equations, which were defined using `\begin{align}<maths>\end{align}`. Note that we also included a label with `\label{<name>}`. This is handy, since it allows us to refer to the equation in our text, eg. from Equation (2) we can see that  $b$  is two more than  $a$ .

# 9 Results

Remember the advice about the previous sections. You need your results to be concise, but once again be concrete, quantitative, and provide enough information that the results could be reproduced and verified. Figures and tables can be very useful. However, while a picture is worth a thousand words, this is not true by itself. Any graph or figure included in a report MUST have:

1. a detailed caption describing exactly what the figure shows (it should almost stand alone);
2. appropriate axes with labels including units; and
3. discussion in the text of the document, not just the caption (make sure you refer to exact figure numbers in the text).

Moreover, they should be easy to read with large enough text, and clearly marked data points. Tables are also good and should follow the same rules. To produce a table, you can perhaps print out a tibble using an R code chunk, or you can type the table using Latex commands like the one in Table 1.

|       |       |       |
|-------|-------|-------|
| cell1 | cell2 | cell3 |
| cell4 | cell5 | cell6 |
| cell7 | cell8 | cell9 |

Table 1: Table to test captions and labels.

A few figures and tables in a document are very useful, but be aware that deluging a reader with figures and tables can be counter-productive. Part of the art of technical writing is choosing good ways of informing the reader of the critical information without diluting it with volumes of irrelevancies.

### IMPORTANT:

- **All of your results must be backed up by some code output.**
  - **Any claims that have no supporting evidence will be ignored!**
- So that means that any claims you make should reference the place in the appendix or some code where you produce the calculations or graphs that you use. You can use a footnote for this or just put a comment in brackets.
  - Here’s how you can insert a footnote<sup>1</sup>.
- Eg. Likewise, the sons of Feanor, Celegorm and Curufin, dwelt in Nargothrond, and their oath to regain the Silmarils would likewise call them. According to our simulation, on average there was space for all these human and elven heroes 97% of the time.<sup>2</sup>
- Eg. He pursued the Orcs and came in secret to their fire, witnessing their captain holding aloft his father’s hand as a trophy, bearing the ring of Felagund. Beren slew the captain and took the hand with the ring, managing to escape before the Orcs could kill him. We also found that the average waiting time experienced by Beren was 5.78 seconds. (See “average waiting time” in Appendix E.2. This calculation was generated by the function `wait_beren()` on lines 57 – 73 of the file `silmarils_fns.R`.)

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<sup>1</sup>Including a footnote

<sup>2</sup>See “percentage of times waiting room exceeded” in Appendix D.4. This calculation was generated by the function `room_prob()` on lines 154 – 165 of the file `silmarils.R`. This function processes the data output from the `elvish_wait()` function on lines 13 – 22 of the same file.

## 10 Discussion

Here you discuss the implications of your results in the context of your research project. In other words, tell the reader what you think the results mean. This section will probably involve a bit of a short summary, as you need to remind the reader of what you are doing so that the results are meaningful.

Since research is never finished, you should discuss what the next steps in the project should be.

## 11 Acknowledgements

This is where should acknowledge any help you got, other than those things listed in the references. So things like:

- Research advice given to you by someone else
- Help provided by an AI agent
  - Some AI help is ok, as long as it doesn't do your work for you and you acknowledge and reference it
  - Ask your supervisor for advice on what AI help is appropriate
- Help given by the Student Advisors

For example, I acknowledge that some of this template was adapted from an old University of Adelaide Latex report template.



## 12 Statistical appendix

This section should contain all the code and output that you need to justify the claims you made above. Only include the code that actually contributes to your Results or Methods sections (maybe also the Background section). We don't need to see every single line of code you wrote, just the important bits that are needed for the reader to be able to obtain your results. Again, this is another balancing act between being too long and irrelevant, or being too short and incomplete (which would open you up to claims of academic misconduct or negligence). Research is a tough business.

Also, don't just paste all the code in here. Use subheadings and text to say what the code is, and what it is used for.

### 12.1 Appendix A: First Appendix

This text is under a subheading.

A code chunk running R:

```
a <- 1
b <- a+2
b
```

```
## [1] 3
```

You can run python from an Rmd file. But make sure you have the `reticulate` package installed first. Then the following code will run:

```
import numpy
x = numpy.arange(0.0, 2.0, 0.5)
y = 1 + numpy.sin(numpy.pi*x)
y
```

```
## array([1., 2., 1., 0.]
```

The following code chunk doesn't actually run, it is just displaying the code. You can use this if you want to include some code which takes too many hours to actually compile. But then you need to say which file you actually used to generate the data (since this code is not actually doing anything).

```
a <- 1
b <- a+2
b
```

#### 12.1.1 Appendix A.1: A subsubsection of Appendix A

This text is under a subsubheading.

### 12.2 Appendix B: The second Appendix

We've now started another Appendix.

### **12.2.1 Appendix B.1: A subsection of Appendix B**

This text is under a subsubheading.

### **12.2.2 Appendix B.2: Another subsection of Appendix B**

This text is under a subsubheading.



Figure 5: T-Rex picture. Source Wikipedia: [https://commons.wikimedia.org/wiki/File:Tyrannosaurus\\_Rex\\_colored.png](https://commons.wikimedia.org/wiki/File:Tyrannosaurus_Rex_colored.png)